

Test-Time Training with Masked Autoencoders: Online Version (E)

An Investigation on ImageNet-C Corruptions

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January 5, 2025

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Motivation and Problem Statement

Problem: Generalization under distribution shifts remains a critical challenge in computer vision.

Traditional supervised methods:

- Assume training and test data come from the same distribution.
- Struggle when facing unseen test distributions (real-world scenarios).

Test-time training (TTT):

- Adapts models *on-the-fly* during deployment.
- Uses *self-supervision* to update model parameters per test input.

Test-Time Training with Masked Autoencoders (TTT-MAE)

TTT-MAE [Gandelsman et al., 2022]:

- Incorporates Masked Autoencoders into TTT.
- Improves robustness to distribution shifts on visual benchmarks.

My Work:

- 1 Evaluate TTT-MAE on **ImageNet-C** [Hendrycks and Dietterich, 2019].
- 2 Analyze failure cases
- 3 Propose an **online variant** of TTT-MAE.

ImageNet-C

- **ImageNet-C** is a benchmark designed for model robustness under *15 corruption types*, each with 5 severity levels.
- Examples: **Brightness, Glass Blur, Motion Blur, Zoom Blur, Contrast**, etc.
- We follow the data augmentation strategy in [Gandelsman et al., 2022] (random crops, flips).

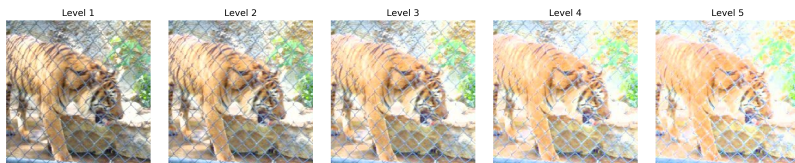


Figure: Example corruption: Brightness, severity levels 1–5.

Model Architecture

- **Y-shaped architecture** [Gandelsman et al., 2022]:
 - Shared **ViT-Large encoder** pre-trained as a MAE.
 - Two heads:
 - 1 A **lightweight decoder** for reconstruction.
 - 2 A **classifier** for the main task.
- **Total parameters:** 416+ million.

Test-Time Training (TTT)

- During testing, we:
 - ① Mask 75% of patches in the test image.
 - ② Optimize mean squared error (MSE) between reconstructed and original patches.
 - ③ **Update only** encoder + self-supervised head for each test input.
 - ④ Classifier head remains *fixed*.
- Predictions are made after these TTT adaptation steps.
- Standard TTT resets weights to pre-trained state for every new test image.

Comparison with Original Paper's Results

Experiment Protocol Modifications (due to resource constraints):

- 1 Evaluated **2 images/category** instead of 50, i.e. 2,000 test images per corruption.
- 2 **Batch size** for gradient updates reduced from 128 to **16**.

Impacts:

- Some corruptions (Brightness, Defocus blur, Fog, Frost) do **not** surpass baseline.
- Others (e.g., Gaussian noise) show **higher** accuracy than reported.
- Highlights the sensitivity to batch size and learning rate.

Quantitative Results Example

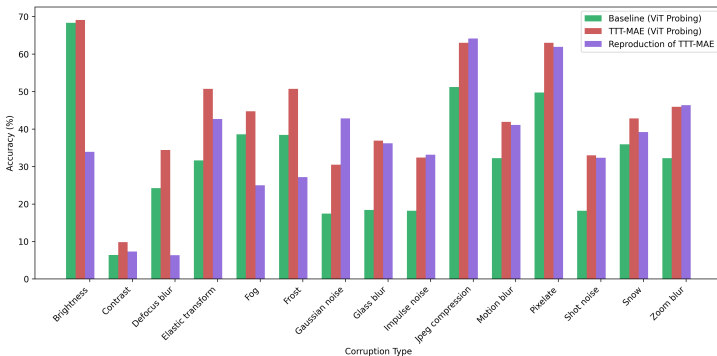
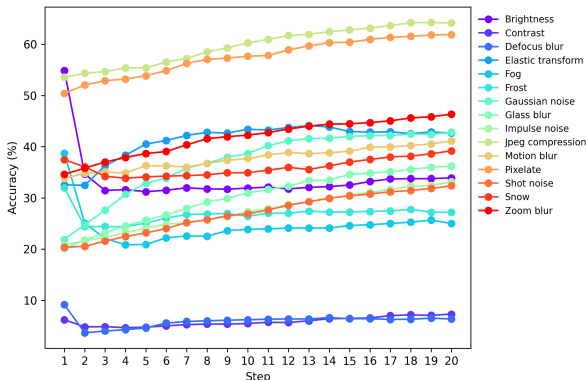


Figure: Accuracy on ImageNet-C (level 5) showing baseline vs. TTT-MAE. Baseline and TTT-MAE results are taken from [Gandelsman et al., 2022]

Accuracy vs. TTT Steps

- Generally, **more TTT steps** improves accuracy [Gandelsman et al., 2022].
- Some corruptions exhibit a **drop** early, likely due to large learning rate and small batch size.



Failure Cases (Defocus Blur)

- *Decoupling* between reconstruction and classification.
- Even when reconstruction loss is stable and near zero, classification can fail.
- Some images remain misclassified throughout all TTT steps (chaotic classification loss).

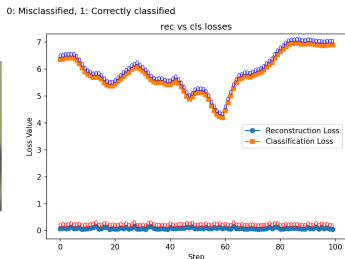
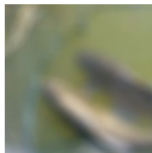


Figure: Reconstruction & classification loss evolution during TTT.

Qualitative Analysis

- Reconstruction quality does **not** necessarily improve with more TTT steps.

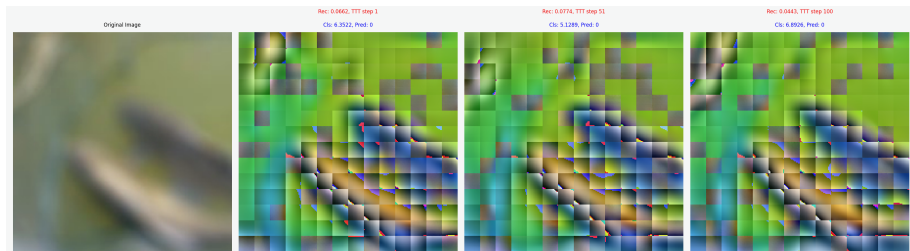
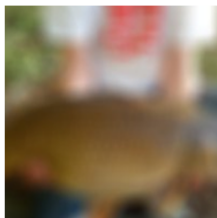


Figure: Decoded reconstructions over TTT steps for *Defocus blur*.

Successful Case



0: Misclassified, 1: Correctly classified

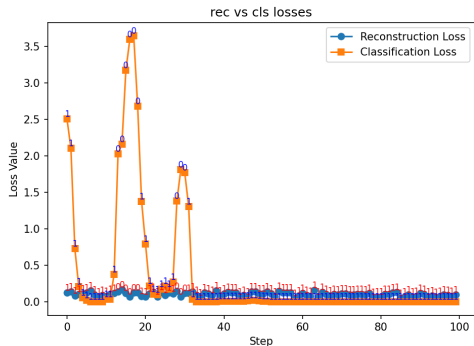


Figure: Reconstruction & classification loss evolution during TTT.

Successful Case

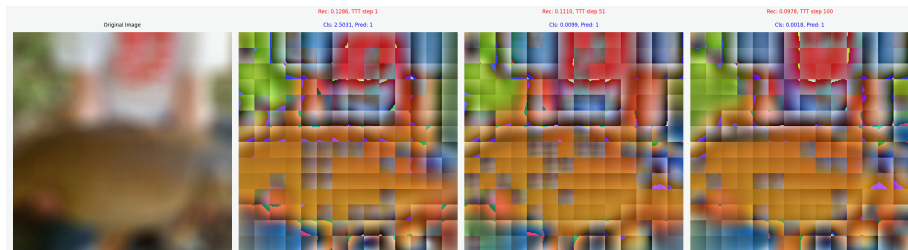


Figure: Decoded reconstructions over TTT steps for *Defocus blur*.

Online TTT-MAE

- **Online variant:** Encoder weights *not reset* between test samples.
- Updates are **cumulative** across samples.
- Evaluate on {Brightness, Defocus blur, Frost} with a batch size of 1, 2 images/category.

Findings:

- Significant **accuracy gain** vs. baseline TTT.
- Gains appear even with **fewer steps**.

Online TTT-MAE: Quantitative Results

Corruption	Baseline TTT (%)	TTT Online (%)
Brightness	69.5	74.1
Defocus blur	26.55	34.15
Frost	43.7	53.95

Table: Accuracy results on ImageNet-C level 5.

Online TTT-MAE: Quantitative Results

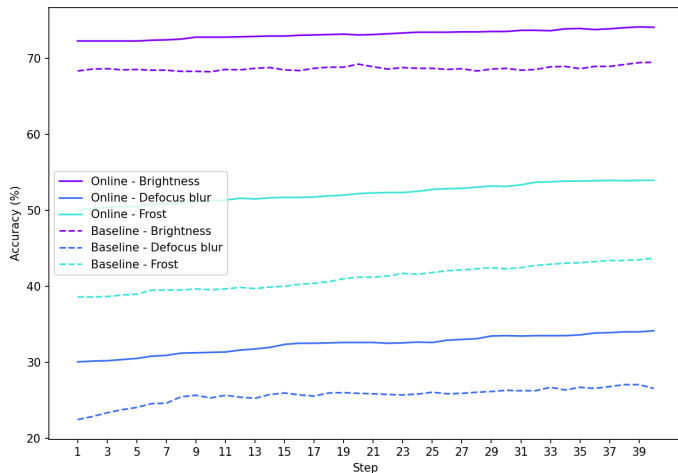


Figure: Accuracy vs. TTT step in the online setting.

Conclusion and Future Work

- **Strengths:**

- Improves performance on most corruptions over baseline.
- Online variant can significantly boost accuracy.

- **Limitations:**

- Large memory/computational overhead (ViT-Large).
- High sensitivity to hyperparameters (learning rate, batch size, number of steps).

- **Future Directions:**

- Further investigate reconstruction-classification coupling.
- Explore different reconstruction losses or bridging tasks.
- Validate on diverse datasets/scenarios.

References I



Gandelsman, Y., Sun, Y., Chen, X., and Efros, A. A. (2022).
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